



IIT Pokerbots 2026

Official Tournament Architecture & Scheduling Logic

0: Rating System Fundamentals

To maintain a high-integrity leaderboard, every bot is tracked using two primary variables inspired by Bayesian rating systems:

- **Skill Estimate (μ):** The system's current "best guess" of your bot's true strength.
- **Uncertainty (σ):** The degree of confidence the system has in that guess. A high σ means we need more data; a low σ means your performance is stable.

The **Leaderboard Score** you see is not a simple average. It is calculated as $\mu - \sigma$. This represents a **Conservative Lower Bound**. Statistically, the system is approx **84% confident** that your actual skill level is at least this high. As you play more matches and σ shrinks, your displayed score naturally "climbs" toward your true μ .

0.1 Dynamic Variance & Code Pushes

We don't believe in full rating resets, as they destroy useful historical data. Instead, we use **Variance Boosting** to keep the competition fluid:

- **New Uploads:** When you push a code update, we keep your μ (a "warm start") but reset your σ to the maximum. This signals to the scheduler that your bot is now an "unknown quantity," triggering a spike in match frequency to find your new rank quickly.
- **Idle Decay:** If your bot stops playing, its σ slowly increases over time. This reflects "temporal decay"—the longer it has been since we saw your bot in action, the less certain we are that your old rating still holds true.

Some participants may consider "resetting" their rating to clear a losing streak. Our architecture makes this unnecessary and counter-productive:

1. **Warm Starts:** By keeping your μ during a code push, we ensure you don't have to "grind" through low-level bots again.
2. **Sigma Spikes:** A new upload automatically gives you the "fresh start" you need by blowing up your σ . This allows for rapid upward mobility if your new code is genuinely better, without losing the baseline progress you've already made.

0.2 How Ratings are Updated

The system does not use a fixed "K-factor" like standard Elo. Instead, every match result ($W | L | D$) is processed through a Bayesian update cycle that accounts for the relative strength of both opponents:

- **Winning vs. a Strong Opponent:** If you win against a bot with a higher μ and lower σ , your rating gains are maximized because you have provided "high-quality" evidence of skill.
- **Losing vs. a Strong Opponent:** Your rating will barely move. The system already expected this outcome, so your "true skill" hasn't been disproven.
- **The "Upset" Penalty:** Losing to a much lower-ranked bot causes a significant drop in μ and a spike in σ . The system now views your performance as "unstable" and will schedule more matches to see if the loss was a fluke or a genuine decline in strategy.

The system is designed to be **rigorous**; it values recent evidence over old history while ensuring that one bad night of testing doesn't permanently ruin your standing.

1: The Continuous Scheduling Engine

The current preliminary phase of the IIT Pokerbots Tournament is powered by a continuous, dynamic scheduling engine that operates 24/7. This system is designed to maximize the rate of meaningful matchups while ensuring fairness and rapid convergence of bot ratings.

It tries to solve the classical “cold start” and “rating stagnation” problems in competitive bot environments.

1.1 The Multi-Objective Priority Function

Every potential pairing is evaluated by a priority score P , ensuring that every match serves a specific purpose:

$$P = W_1\sigma_{\text{sum}} + W_2e^{-\frac{|\Delta\mu|}{\beta}} + W_3S_{\text{sum}} - P_{\text{penalty}}$$

- **Active Variance Exploration (W_1):** The system prioritizes bots with high σ (uncertainty). This ensures that new entrants or updated codebases receive a high density of matches to reach their true μ (rating) with minimal delay.
- **Skill Proximity (W_2):** The scheduler tries to pair you with opponents at a similar level. Playing against someone way better or way worse doesn’t tell us much about your actual skill, so matching you with “peers” keeps your rating updates accurate and prevents “one-sided stomps” that don’t provide useful data.
- **Starvation & SLA Compliance (W_3):** We implement a strict starvation threshold. Any bot remaining idle for longer than 25 minutes receives a flat +1000.0 priority boost, guaranteeing it a seat at the next available table.
- **Repeated-Pairing Decay (P_{penalty}):** To prevent bots from being played with only a few singular opponents, we apply an exponentially decaying penalty to recent pairings. This forces the scheduler to explore a wider variety of matchups, improving the overall robustness of the rating system.

1.2 Structural Integrity Mechanisms

To ensure the leaderboard remains fluid and accurate, the engine employs two critical “Safety Valves”:

- **The Sigma Floor:** We maintain a hard floor (σ_{floor}) and high volatility constant (τ). This prevents ratings from “freezing” after many games, ensuring that a late-tournament breakthrough in your code is immediately reflected in the ranks.

The hard floor ensures the variance does not become arbitrarily small, which would otherwise lead to a “rich get richer” scenario where top bots only play each other.

The high volatility constant ensures that even bots with many games under their belt can still experience significant rating changes if they suddenly improve, preventing the leaderboard from becoming stale.

- **Neighborhood Exploration:** 20% of all match slots are reserved for “cross-neighborhood” matches. This breaks up local skill aggregates and matches bots from different rating “clusters” against each other. This promotes a more accurate global ranking and prevents the formation of isolated “skill islands” where certain bots only play a small subset of the field.

2: The Finals - Deterministic Round Robin

The tournament concludes with an exhaustive $nC2$ phase where scheduling heuristics are replaced by absolute head-to-head performance.

2.1 Full Combination Play

Every bot will play every other bot. This phase eliminates all scheduling variance, providing a transparent and definitive ranking based on direct competition.

2.2 Variance-Dampened Scoring

Final standings are determined by a Sigmoid-Weighted Outcome system. Rather than summing raw chip counts; which can be heavily skewed by a single high-variance “all-in” hand. Hence match results are passed through a non-linear mapping.

This weighting ensures that:

- Consistent, incremental wins are properly rewarded.
- Extreme outliers and “lucky” statistical anomalies are dampened.
- Extremely small wins are attributed less than a full point, reflecting the inherent uncertainty in very close matches.
- The final leaderboard reflects a bot’s fundamental edge over the field rather than its proximity to high-variance events.

Note on Convergence: The weights (W_1, W_2, W_3) are mathematically tuned so that the preliminary leaderboard acts as a high-speed proxy for the final Round Robin. By the time the qualifiers end, the ratings are designed to have converged closely enough that the $nC2$ finals serve as a definitive verification of the top-tier rankings rather than a total reshuffle.